Logarithms

1 Rewrite each of the following equations in logarithmic form:

a $5^2 = 25$ **b** $3^x = 81$ **c** $3^1 = 3$ **d** $2^0 = 1$

e $4.4^0 = 1$ **f** $25^{1.5} = 125$ **g** $4^{\frac{5}{2}} = 32$ **h** $4^{-2} = 0.0625$

i $4^{-3} = \dfrac{1}{64}$ **j** $x^{1.5} = 64$ **k** $4^m = 9$ **l** $5^m = q$

m $w = h^q$ **n** $\dfrac{1}{m^{-j}} = n$

2 Rewrite each of the following equations in logarithmic form:

a $10^2 = 100$ **b** $10^{-2} = \dfrac{1}{100}$ **c** $10^{\frac{1}{2}} = \sqrt{10}$ **d** $10^{-\frac{1}{2}} = \dfrac{1}{\sqrt{10}}$

3 Rewrite each of the following equations in exponential form:

a $\log_4 16 = 2$ **b** $\log_5 5 = 1$ **c** $\log_8 1 = 0$ **d** $\log_2 0.125 = -3$

e $\log_3 \dfrac{1}{3} = -1$ **f** $\log_{5.8} 33.64 = 2$ **g** $\log_{\frac{1}{3}} 9 = -2$ **h** $\log_x 32 = 5$

i $\log_8 x = 6$ **j** $\log_g 5 = 3$ **k** $\log_7 m = 40$ **l** $\log_y k = u$

m $\log_r p = y$

4 Rewrite each of the following equations in exponential form:

a $\log_{10} 10\,000 = 4$ **b** $\log_{10} \left(\sqrt{10}\right) = \dfrac{1}{2}$

c $\log \left(\dfrac{1}{10\,000}\right) = -4$ **d** $\log \left(\dfrac{1}{\sqrt{10}}\right) = -\dfrac{1}{2}$

5 True or false: The equation $4\,(1.09)^x = 20$ is equivalent to $x = \log_{1.09} 5$.

Evaluating logarithms

6 For each of the following equations:

i Rewrite the equation in logarithmic form.

ii Find the value of x, correct to two decimal places.

a $10^x = 820$ **b** $10^x = 0.0002$ **c** $10^x = \dfrac{13}{50}$ **d** $10^x = 14\,000$

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7 For each of the following numbers:

i Express the number as a power of 10.

ii Hence state the base ten logarithm of the number.

a $1\,000\,000$ **b** $0.000\,001$ **c** 1 **d** $\frac{1}{10}$

8 Evaluate the following logarithmic expressions without the use of technology:

a $\log_2 64$ **b** $\log_2\left(\frac{1}{2}\right)$ **c** $\log_5 0.04$ **d** $\log_7 1$

e $\log_{1.1} 1.21$ **f** $\log_{0.5} 4$ **g** $\log_{\frac{1}{5}} 25$ **h** $\log_{27} 3$

i $\log_2\left(\frac{1}{16}\right)$

9

a Find the value of the following:

i 5^0 ii $\left(\frac{3}{4}\right)^0$ iii 12^0 iv 0.5^0

b The logarithm, with any base, of what number is equal to zero?

10 In order to estimate $\log_{10} 90$ without a calculator:

a Find $\log_{10} 10$.

b Find $\log_{10} 100$.

c Between which two consecutive integers would you expect $\log_{10} 90$ to lie?

11 For each of the following common logarithms:

i Between which two consecutive integers would you expect the logarithm to lie?

ii Find the value of the logarithm, correct to four decimal places.

a $\log 8892$ **b** $\log 0.053$ **c** $\log 534$ **d** $\log 0.776$

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12 Evaluate the following logarithmic expressions, correct to two decimal places:

a $\log_{10} 40$

b $\log_{10} 7692$

c $\log_{10} 0.005\,79$

d $\log_{10} 8$

e $\log_3 8$

f $\log_6 20$

g $\log_{0.5} 7$

h $\log_{0.3} 15$

i $-\log_2 23$

j $5 \log_4 17$

k $\log_5 \left(20^2 \times \dfrac{3}{43}\right)$

l $\log_3 6 + 2 \log_3 10 - 5 \log_3 15$

13 Find the number such that the base ten logarithm of the number is 2.5. Round your answer to two decimal places.

14 Find the number such that the base ten logarithm of the number is -1.2. Round your answer to three decimal places.

15 Find the number, if the logarithm of the number is 0.

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